

CAD Viewer — User Manual

Version: v0.54 · **URL:** `http://localhost:8088/`

CAD Viewer is a real-time collaborative browser app for viewing CAD geometry (GLBs from SolidEdge, FreeCAD, Fusion 360, OpenCascade..., plus STEP, IGES and OBJ files converted on the server). Create a session, share a model, and everyone on your LAN follows your camera and part visibility live — or just use it as a plain local viewer.

1. Starting the app

Portable zip (Windows, no installs):

1. Unzip `cad-viewer-portable.zip` anywhere.
2. Double-click `start.bat`. It launches the server and opens the browser.
3. The app is served at **`http://localhost:8088/`**

From source: `npm start` (or `node src/server.js`) in the project folder, then open the same URL.

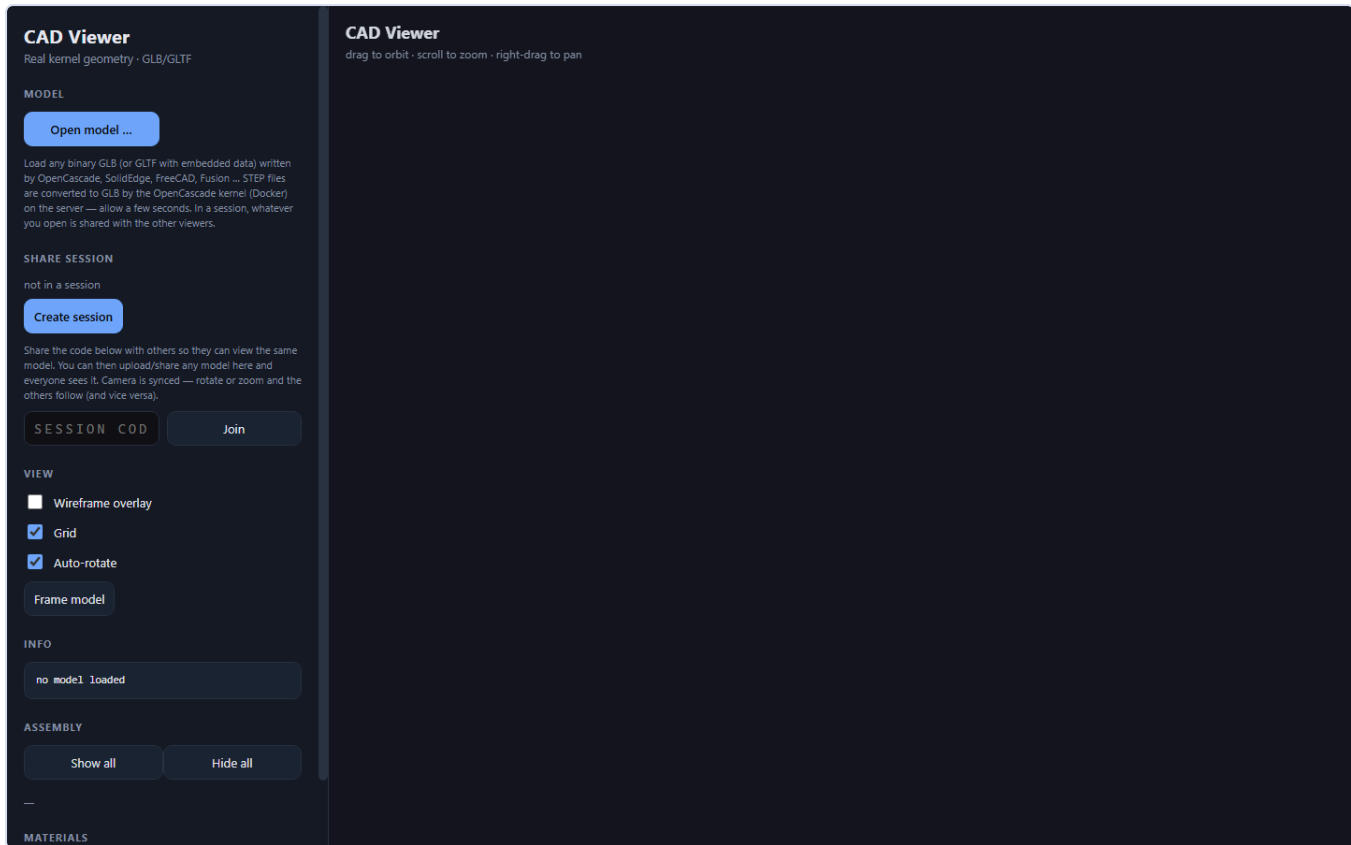
Changing the port

The default port is **8088**. Any of these overrides it:

Method	Example
Launcher argument (zip)	<code>start.bat 5555</code>
<code>port.txt</code> file next to the launcher (<code>start.bat</code> / <code>start.sh</code>)	create <code>port.txt</code> containing <code>5555</code>
Environment variable	<code>set PORT=5555</code> then <code>npm start</code> (Windows) / <code>PORT=5555 npm start</code> (macOS/Linux)

Precedence: command-line argument → `port.txt` → env `PORT` → default 8088. The join link and `/ip` address always reflect the actual port, so guests don't need to know it.

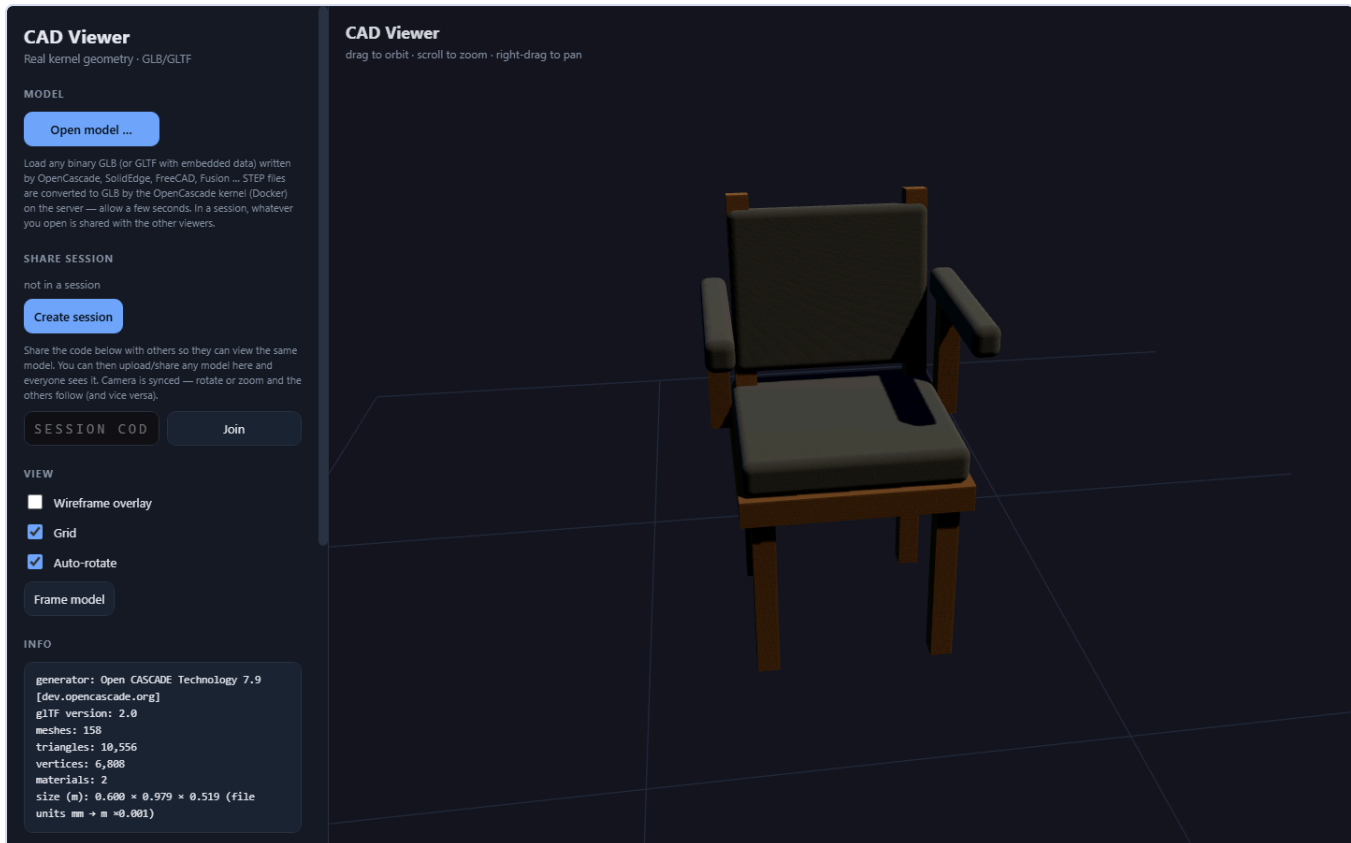
STEP, IGES and OBJ files (.step/.stp, .igs/.iges, .obj) additionally need Docker + the `chair-cq:local` OpenCascade image. Run `install-docker-opencascade.bat` once to set that up. GLB/GLTF files work without any of it.



2. Opening a model

1. Click **Open model...** in the left sidebar.
2. Choose a `.glb`, `.gltf`, `.step`, `.stp`, `.igs`, `.iges` or `.obj` file from your computer.

GLB/GLTF loads instantly. STEP, IGES and OBJ files are converted to GLB by the OpenCascade kernel on the server (a blocking overlay shows progress — a few seconds).

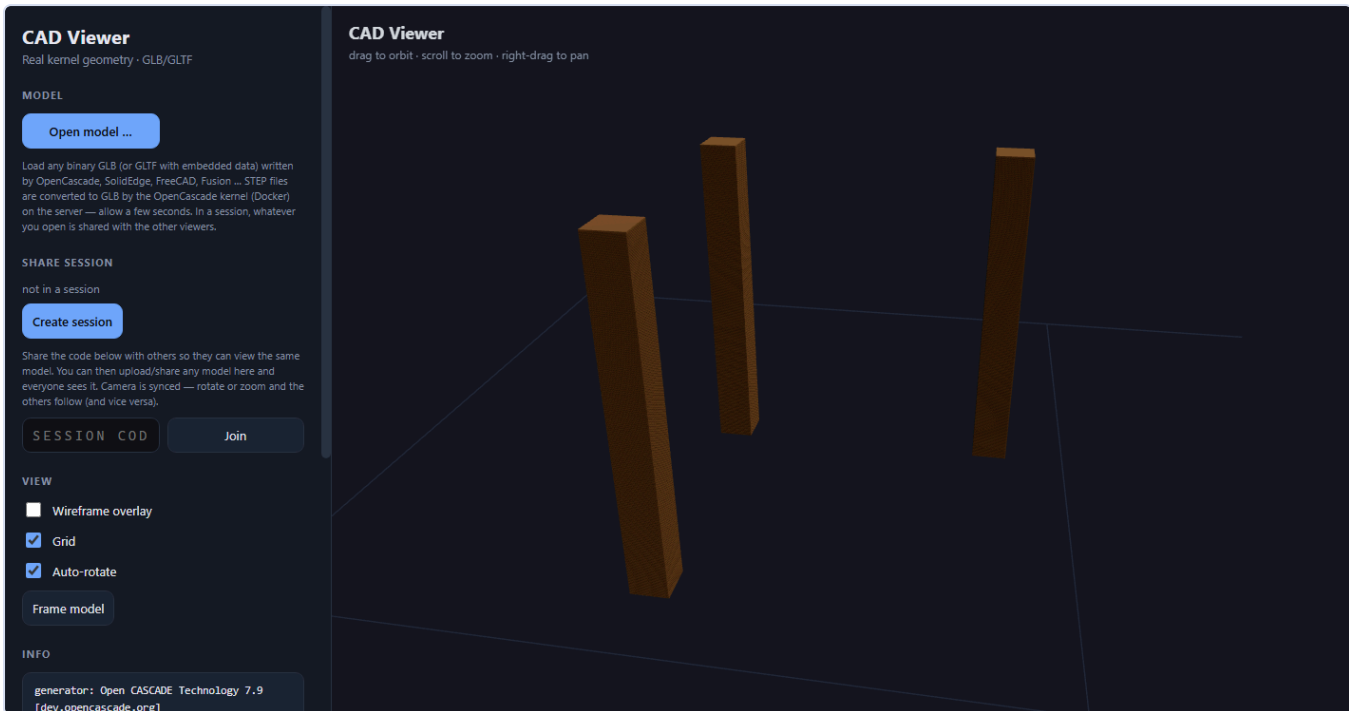


The sidebar shows you:

- **Info** — generator, mesh/triangle counts, dimensions and units
- **Assembly** — the part tree (see §3)
- **Materials** — swatches of every colour in the model

3. The Assembly panel — parts and visibility

Every part of the model appears in the **Assembly** tree. The tree opens expanded to **level 2** (the top assembly plus its direct children); deeper levels are collapsed until you expand them.



Expand / collapse

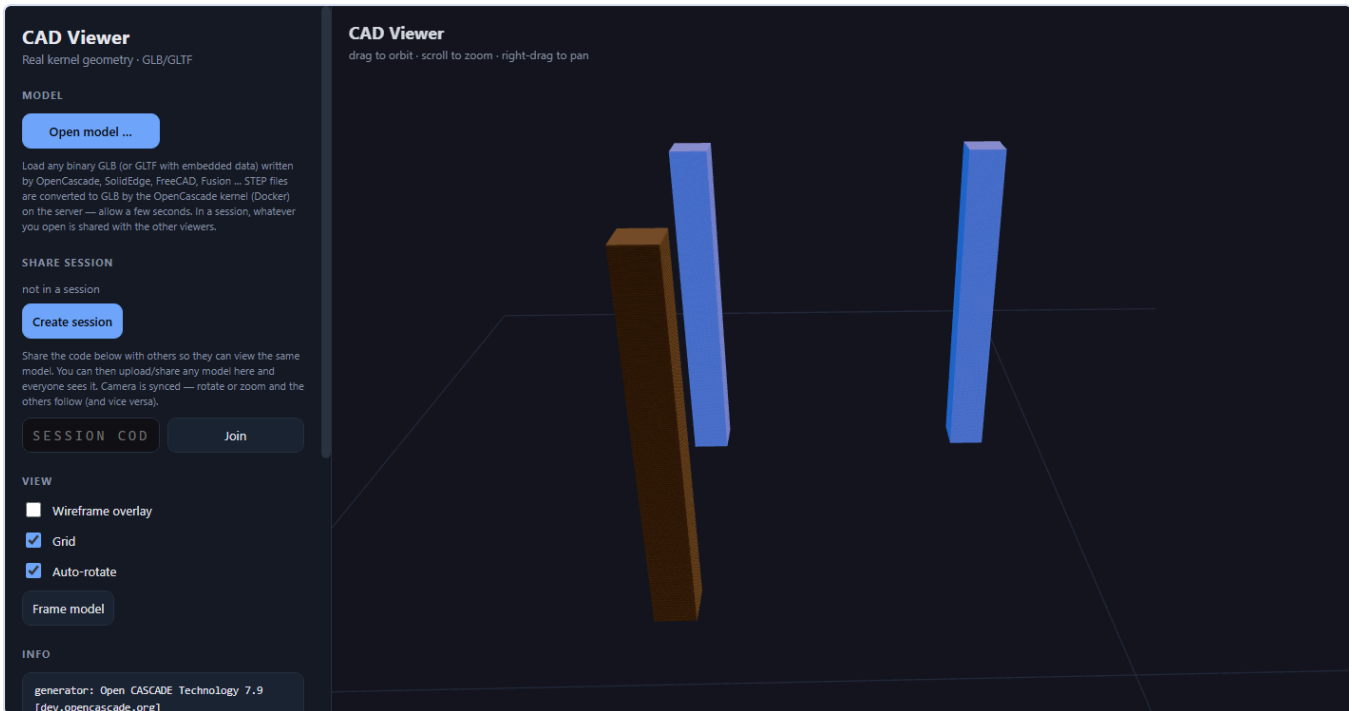
- Rows that contain sub-parts show a **+/- toggle** on the left.
- Click **+** to expand (show children), **-** to collapse (hide them).
- Expanding/collapsing is purely visual — it never changes part visibility.

Show / hide parts

- **Checkbox** per part: uncheck to hide it in the viewport, check to show.
- Toggling a parent cascades to its whole subtree — turning a sub-assembly on brings all its children back with it (three.js needs the whole chain visible to render anything).
- **Show all / Hide all** buttons apply to the whole model at once.

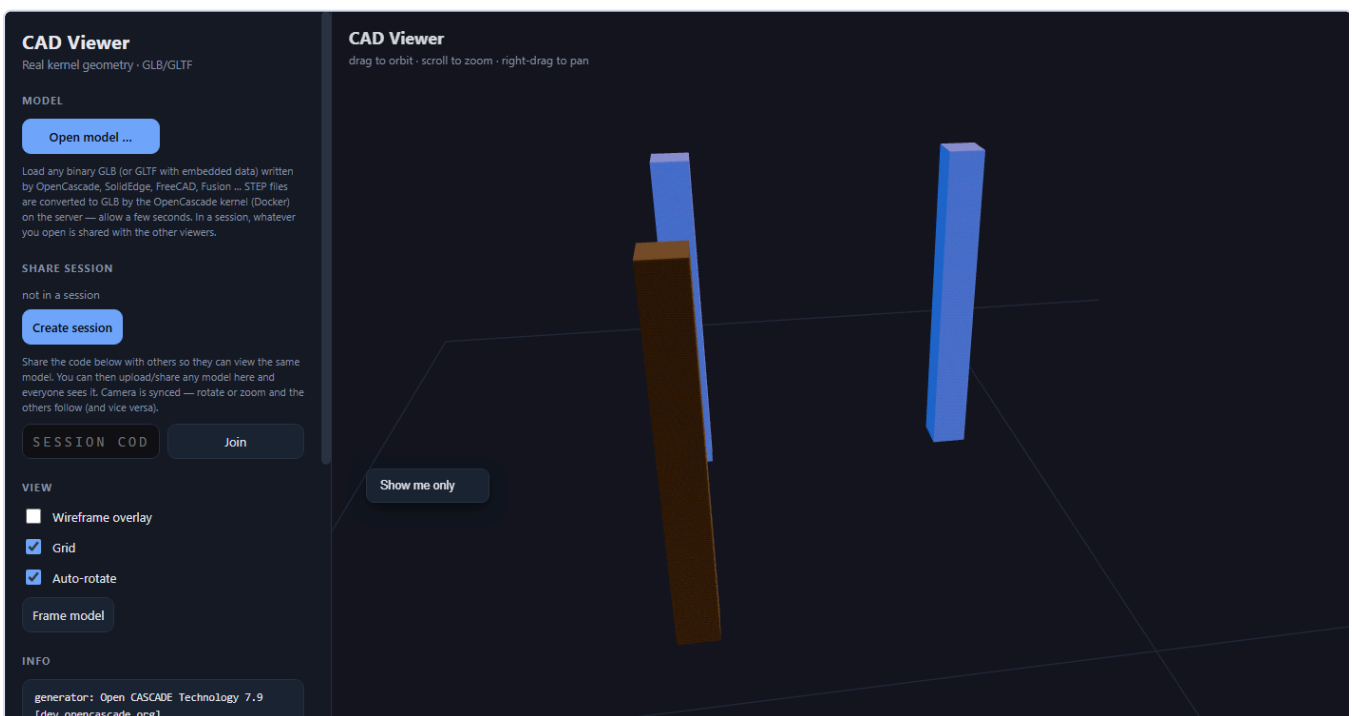
Highlight a part (click the name)

Click a part's **name** to highlight it in blue in the viewport. Click another part to move the highlight; click the same part again to clear it. The highlight uses per-mesh material copies, so other parts sharing the same source material are unaffected.



"Show me only" (right-click)

Right-click a part and choose **Show me only** to hide everything except that part and its children. Ancestors stay visible so the isolated part still renders.



Move a part (X / Y / Z + drag)

With a part highlighted, tick **Move part**, press **X**, **Y** or **Z** to choose an axis, then **left-drag** in the viewport to slide that part along the axis. The highlighted **part moves by itself** — its siblings stay put. An X/Y/Z axis gizmo (red/green/blue) appears at the part so you can see the move directions; the chosen axis glows brighter. The gizmo stays about **1/8 of the screen** at any zoom level — zoom in and it shrinks to

keep that size. **Undo** steps back through your last part movements (up to 200), and **Reset** returns all parts to their original positions and clears the move history. Moves are shared with the other viewers in the session.

In a session (see §5), all of this syncs to every member — show/hide, expand/collapse, and the selection highlight are mirrored live in both directions.

4. Navigating the viewport

Action	Effect
Drag (left button)	Orbit the camera
Scroll	Zoom
Right-drag	Pan
Frame model button	Reset the view to fit the model
Wireframe overlay	Toggle wireframe on all parts
Grid	Toggle the ground grid
Auto-rotate	Slow turntable spin (disabled in a session)

5. Lighting & animation

Lighting

A **Lighting** section in the sidebar lets you tune the light on the model:

Control	What it does	Range	Default
Ambient	Even fill from the sky and ground	0–2	0.9
Key	Main directional light (casts shadows)	0–6	2.4
Fill	Cool fill light from the opposite side	0–3	0.6
Front	Light straight from the viewer's side — handy when the front face is too dark	0–3	0 (off)

Drag a slider and the model updates live. **Reset lighting** restores the defaults.

Animation

If the GLB you load contains **keyframe animation** (e.g. from Blender or a game pipeline — CAD-converted STEP/IGES/OBJ files have none), an **Animation** section appears with:

Control	What it does
Play / Pause	Start or stop the clip
Loop	Repeat the clip (off = play once and stop)
Speed	Playback speed 0.25×–4×
Clip	Pick a clip when the model has several

Session sync

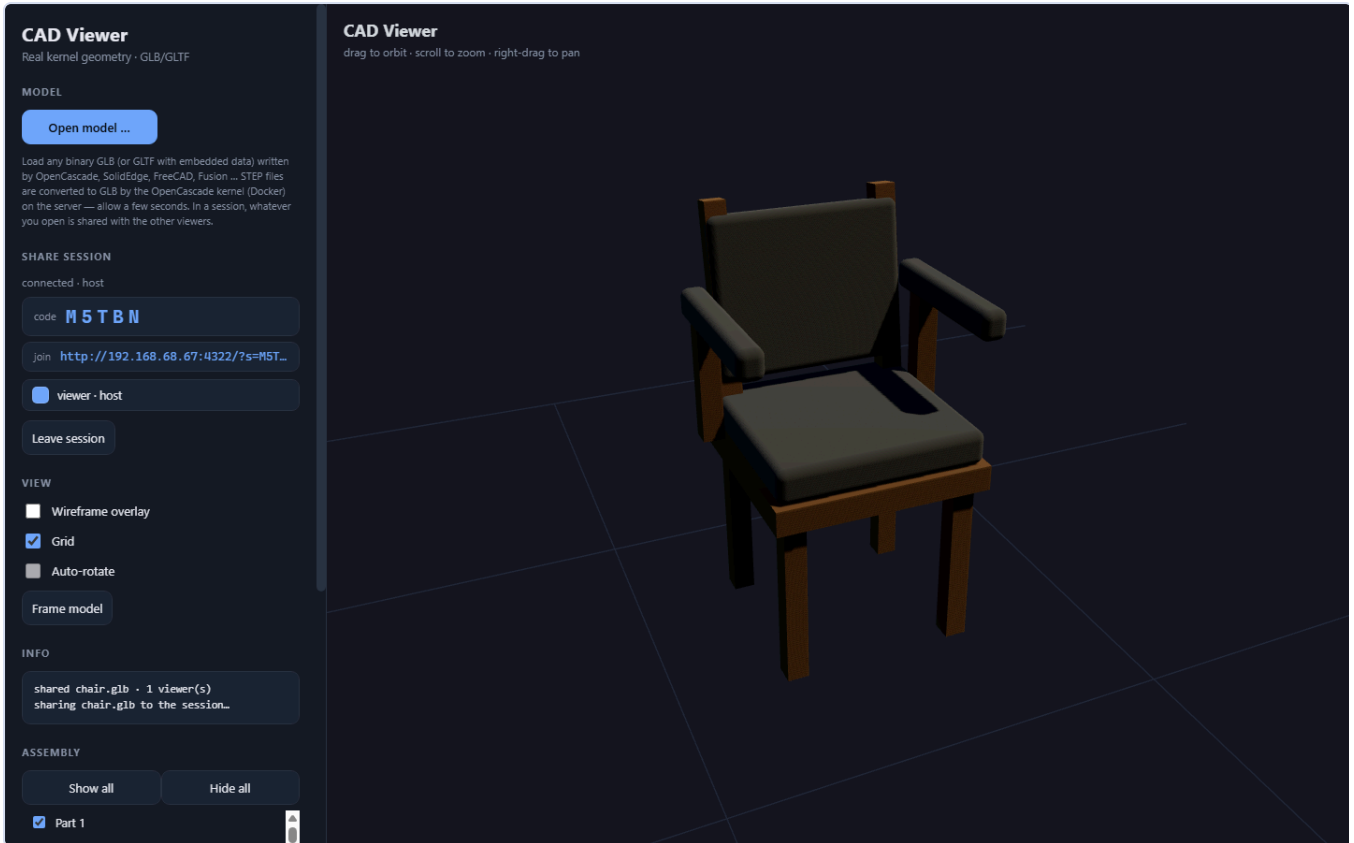
In a session, the **lighting levels (Ambient/Key/Fill/Front)** and the **animation state (clip, play/pause, loop, speed)** are shared with the other viewers, and late joiners receive the current settings. Camera, part visibility, tree expand/collapse and selection sync as well (see §6).

6. Collaborative sessions (host)

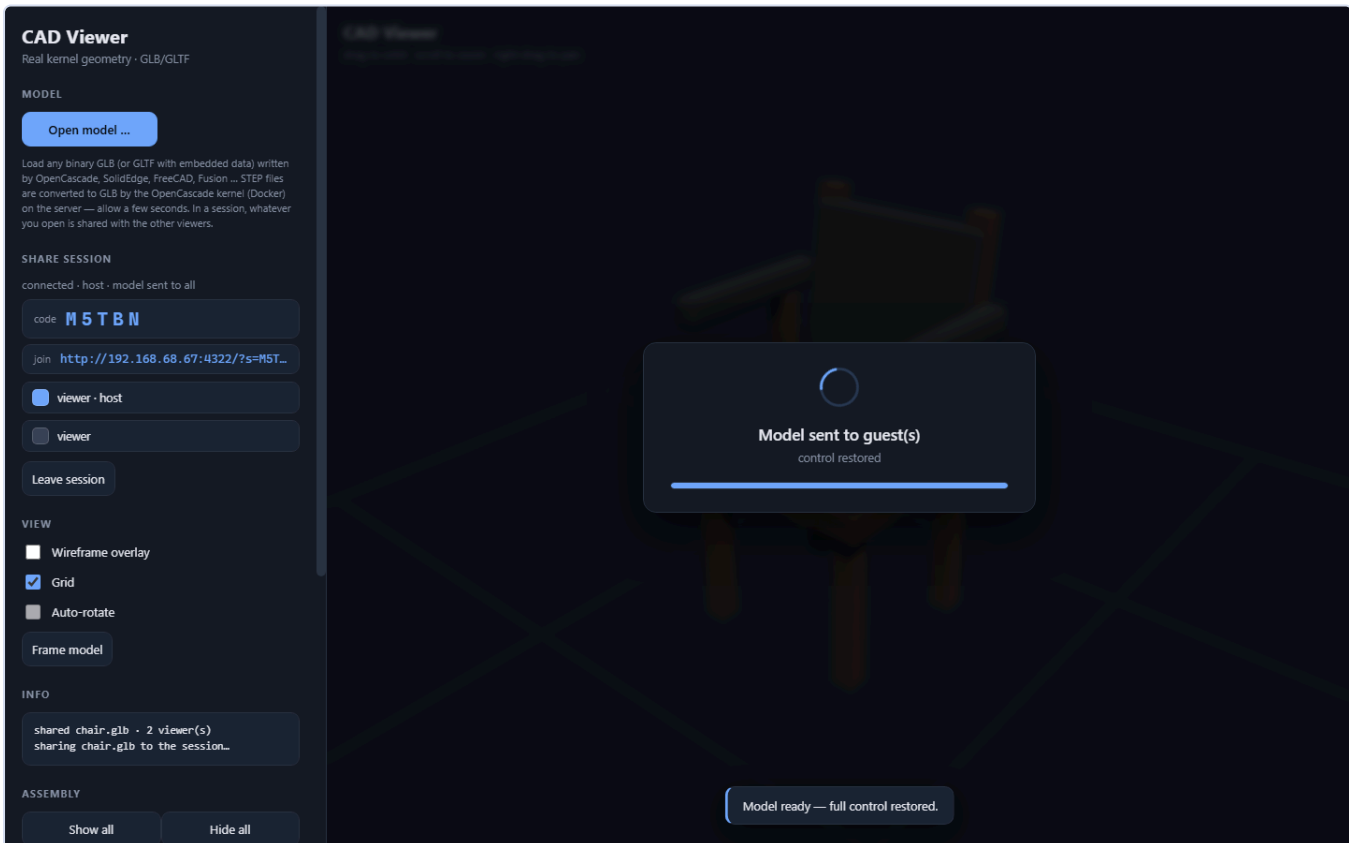
Sessions let other people on your LAN view the same model and follow your camera and part visibility.

On launch you're asked to **enter your name**; it is shown to the other viewers in the roster (e.g. Alice · host , Bob (you)). Your name is remembered for next time.

1. Click **Create session** — a 5-character code appears.
2. **Open a model** (or re-share one you've already loaded).
3. Share the **join link** shown under the code, or the code itself, with the other users.

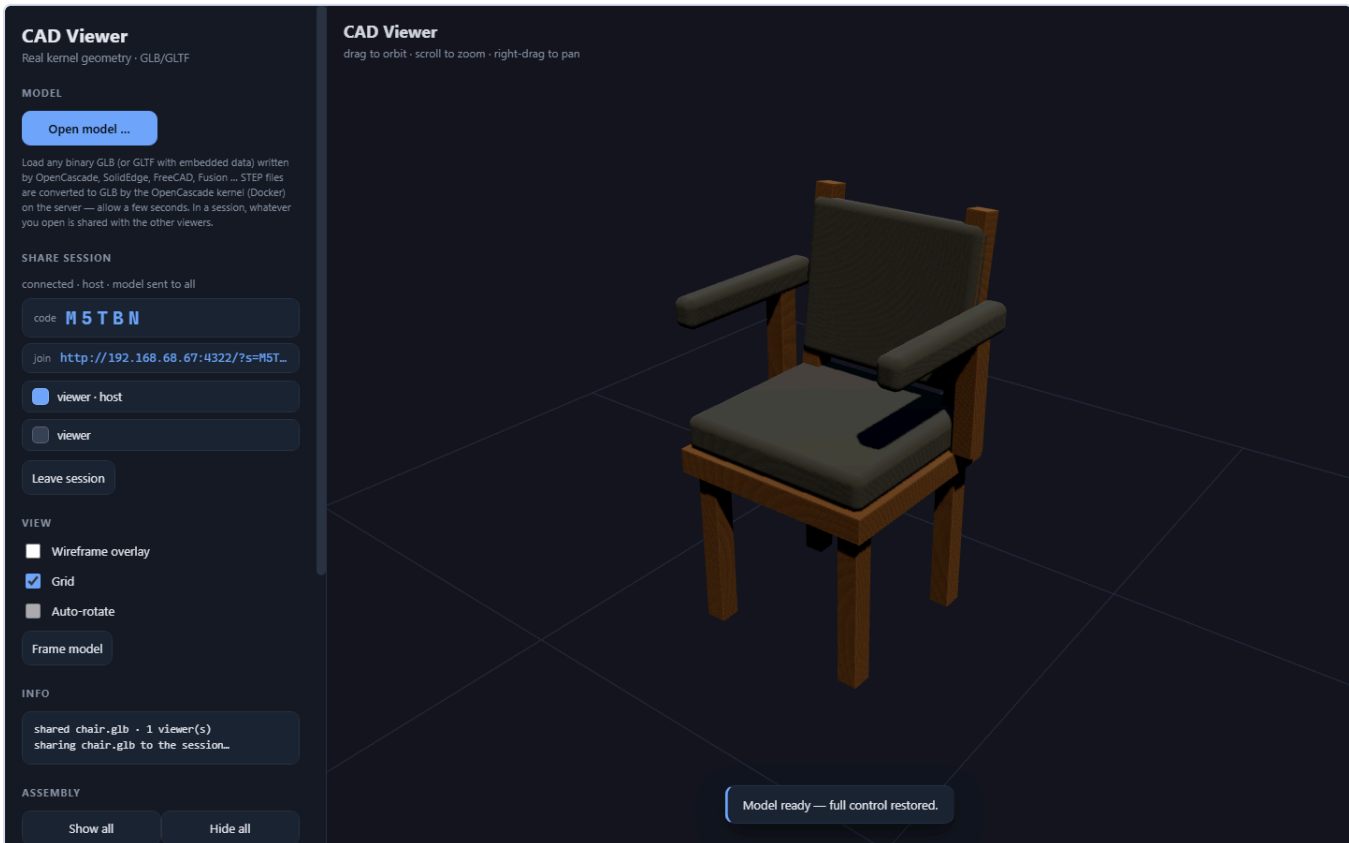


While the model is being handed to guests you'll see a blocking **"Sending model to guest(s)..."** overlay; control is restored automatically once every connected guest confirms receipt.



The roster under the code shows who is connected. The **host** can remove a viewer by clicking **kick** next to their name; the removed viewer's screen is cleared and they see *"removed by host"*. A small status dot

next to "Share session" shows whether the server is reachable (green = up, red = down).

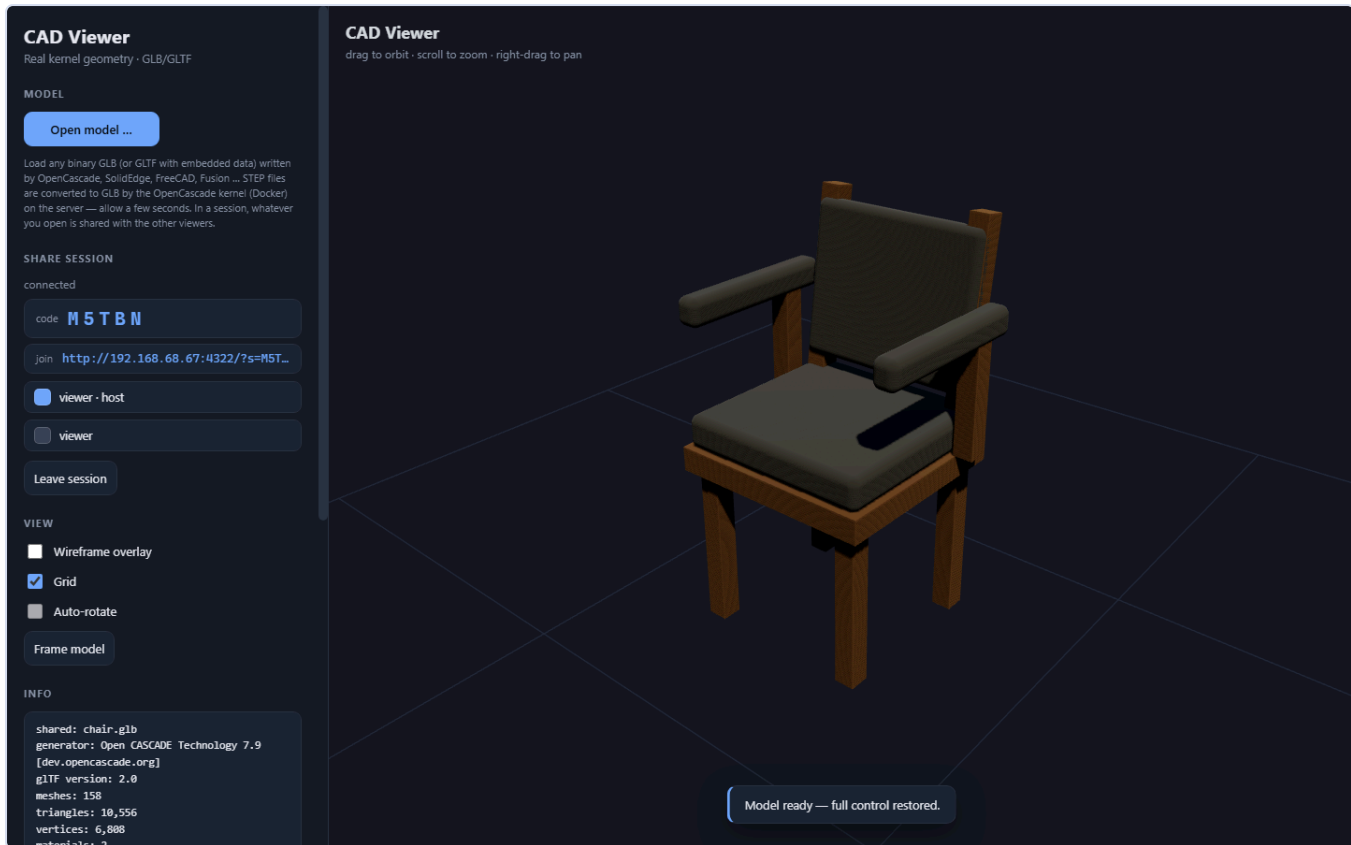


7. Joining a session (guest)

On another computer (same network):

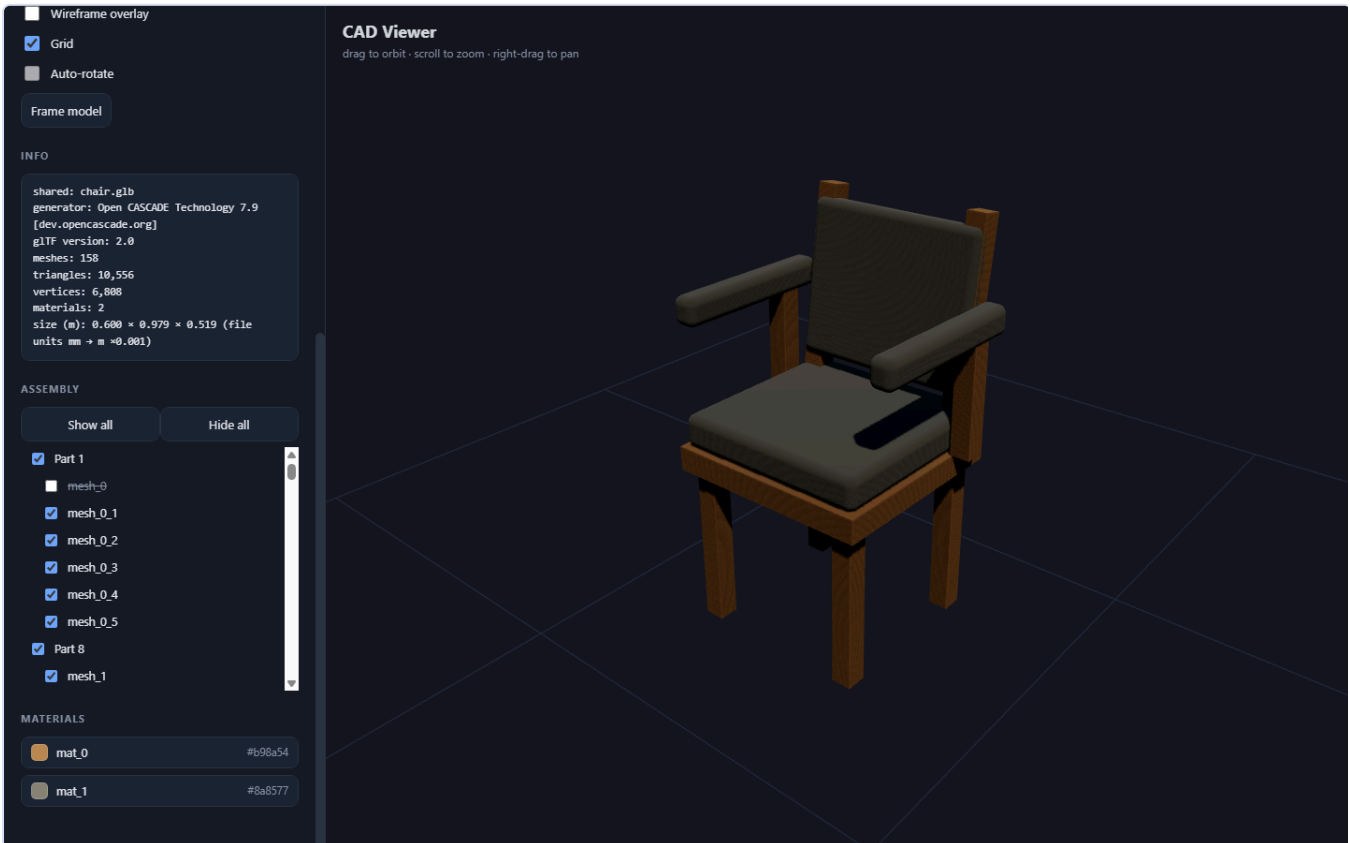
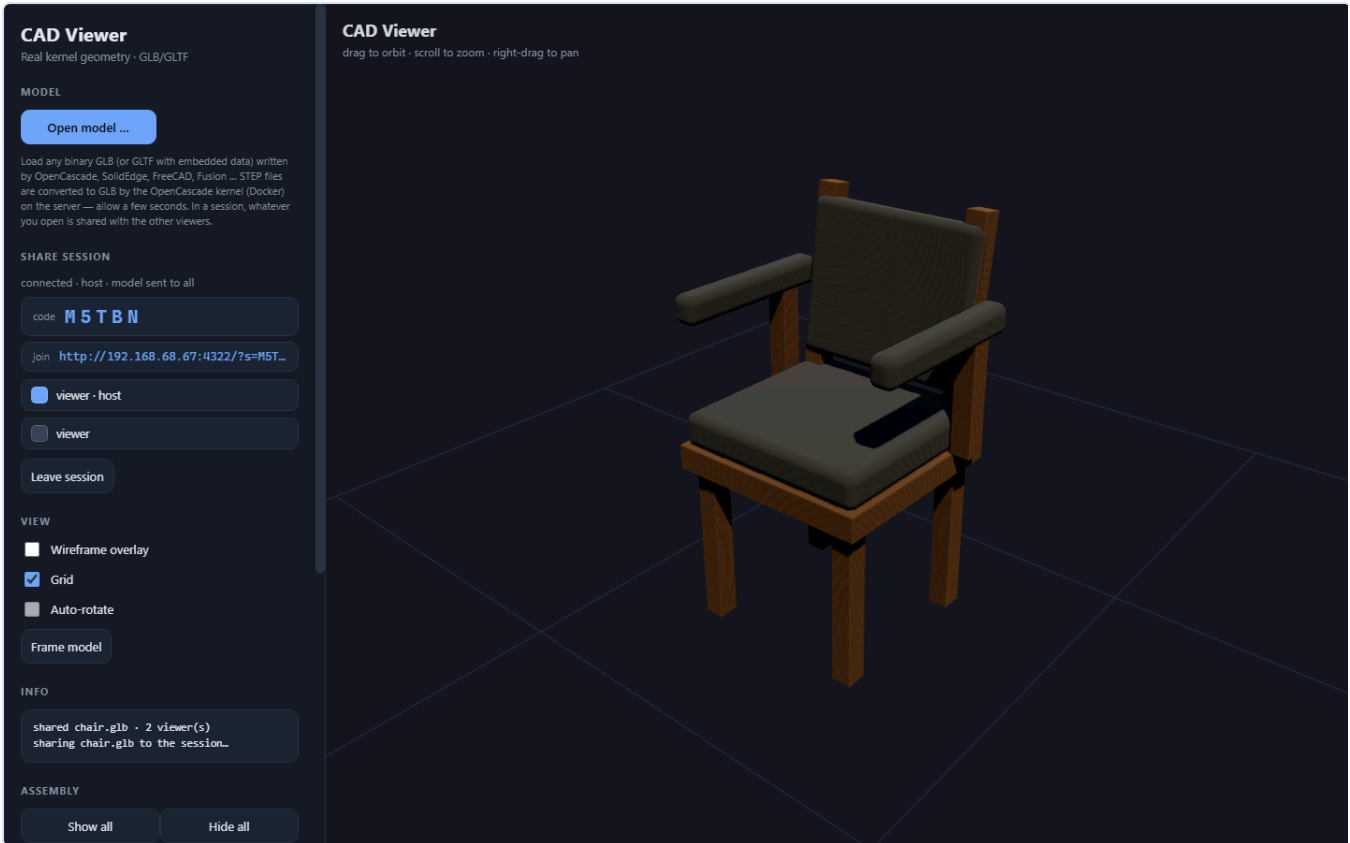
1. Open the **join link** the host sent you, **or**
2. Open the app, type the code into **session code** and click **Join**.

You'll see a **"Receiving model..."** overlay with byte progress, then the model appears — same view, same parts, same visibility as the host.



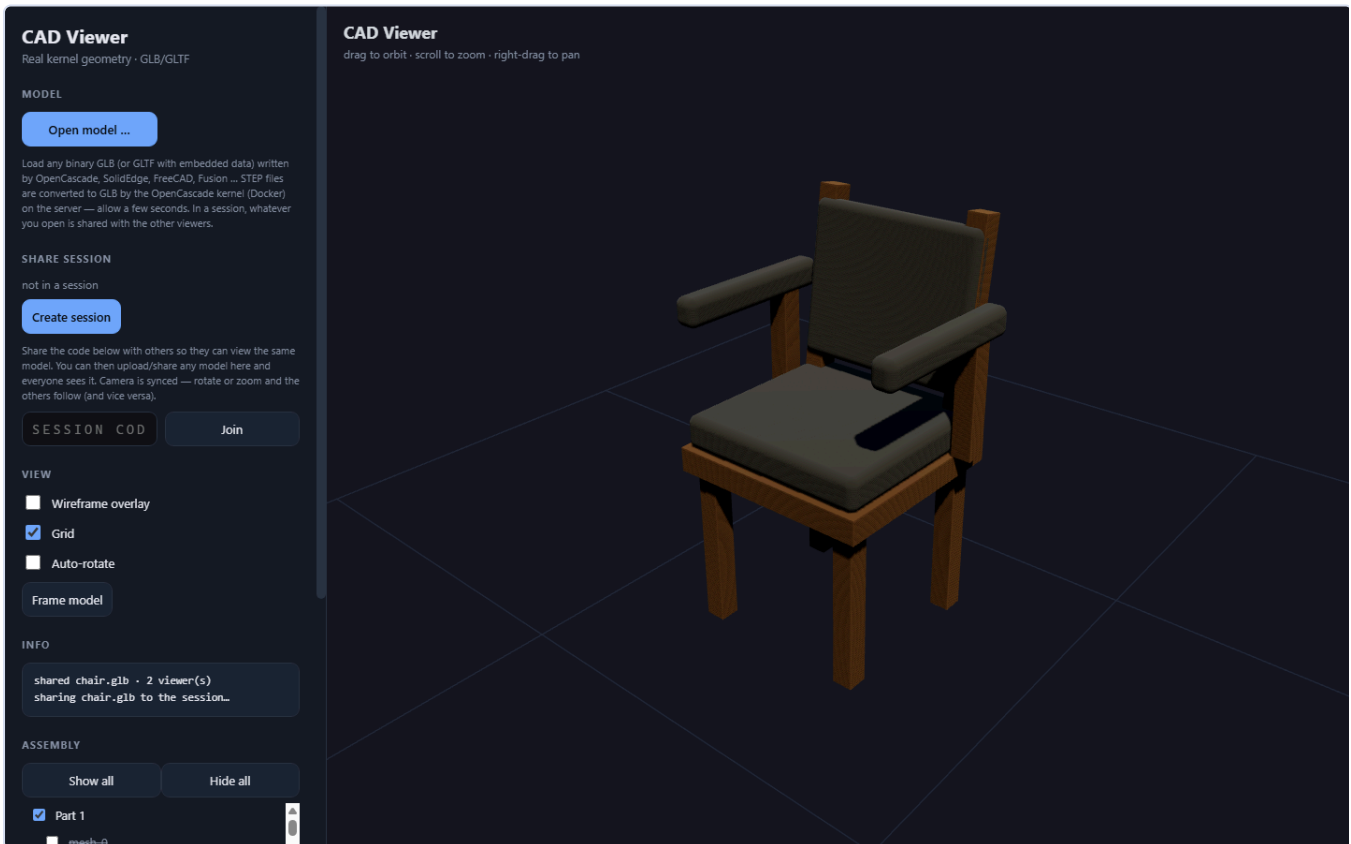
What stays in sync

- **Camera** — whoever orbits/zooms/pans, everyone follows (both directions)
- **Part visibility** — show/hide any part and it changes for all members
- **The model itself** — late joiners automatically receive the current model and visibility state
- **Expand/collapse & selection** — the assembly-tree view and part highlight are mirrored live across members



8. Leaving a session

Click **Leave session** — you keep your local model and return to solo viewing. The session itself stays alive on the server for other members.

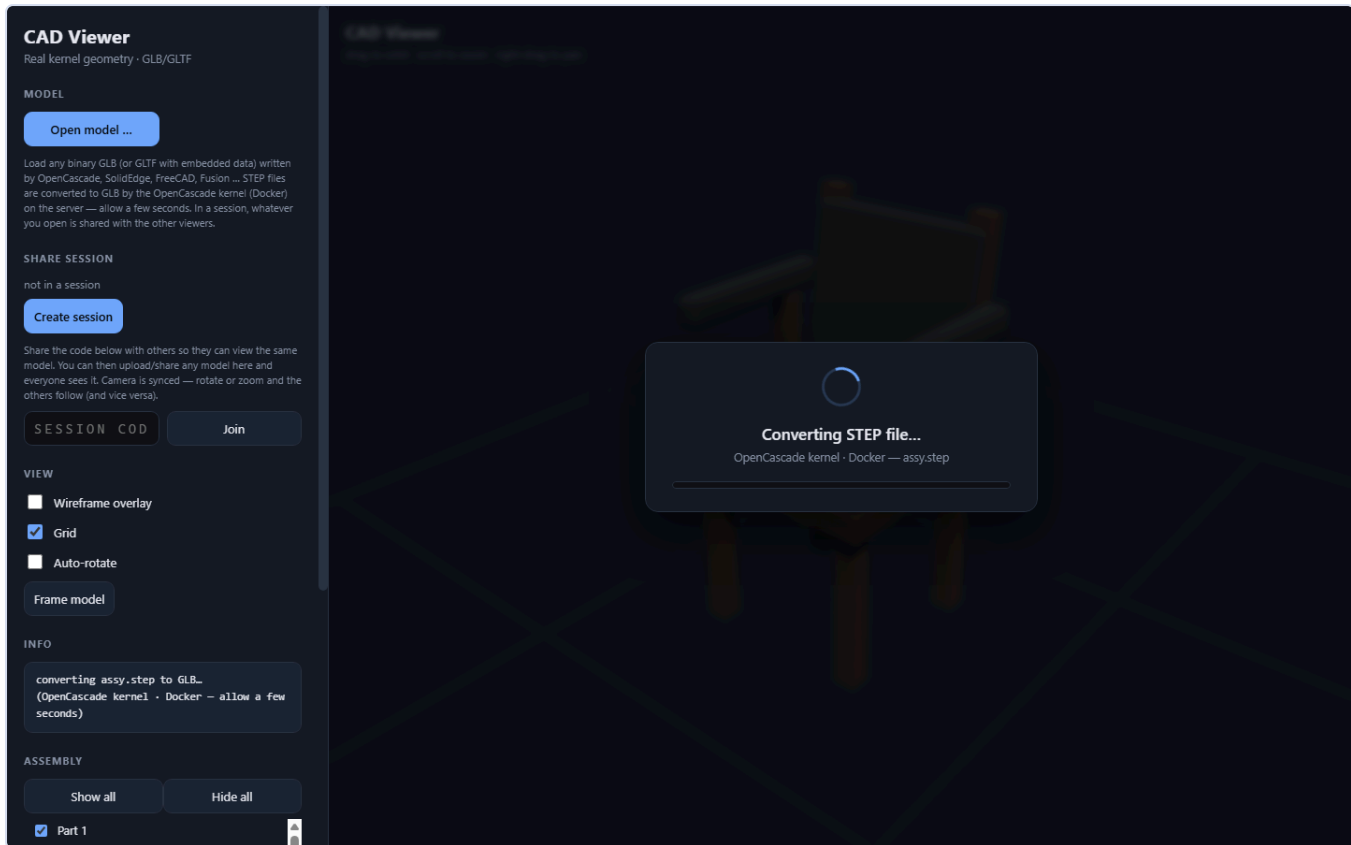


9. Opening a STEP / IGES / OBJ file

STEP, IGES and OBJ conversion happens through the OpenCascade kernel (Docker). The first time you open one you'll see the conversion overlay; when it finishes the GLB is loaded — with the original **colours and materials** preserved, and part names appearing in the Assembly tree.

OBJ colours need the .mtl file too. A Wavefront `.obj` stores colours in a sibling `.mtl` (e.g. `Asm1.obj` + `asm1.mtl`). Select **both** files in the Open dialog (Ctrl+click on Windows) so the materials are applied; opening just the `.obj` still works, but parts get a neutral grey.

OBJ part names. Unlike STEP/IGES (which embed B-rep solid names), OBJ has no mandatory part number. Part names come from the file's `o` (object) / `g` (group) lines when the CAD tool writes them — e.g. SolidEdge's `Asm1.obj` has none, so parts are named `Part1`, `Part2`. If an exporter writes `o Part2` / `o Part3`, those names are used directly.



If Docker or the `chair-cq:local` image isn't installed, these formats show a conversion error while GLB/GLTF continues to work normally.

10. Troubleshooting

Problem	Fix
Can't open the app on another PC	Use the join link (LAN address), make sure both PCs are on the same network, and that port 8088 isn't blocked by a firewall
STEP/IGES/OBJ shows "conversion failed"	Run <code>install-docker-opencascade.bat</code> ; confirm Docker is running with the <code>chair-cq:local</code> image
OBJ parts are grey (no colours)	The <code>.obj</code> was opened without its <code>.mtl</code> — select both files together in the Open dialog
Guest doesn't get the model	The host must be connected with a model loaded — joining an empty session shows nothing until the host shares
Port already in use	Change it: <code>start.bat 4323</code> (zip), a <code>port.txt</code> file, or <code>set PORT=4323</code>
Join link shows the wrong IP	The link uses the host's LAN address; refresh/re-create the session to re-detect it

CAD Viewer v0.54 — collaborative CAD viewing for the LAN.